Voltage Sag Characteristics in Power Distribution Systems under Fault Conditions

Mr. Anmol Saxena[1], Dr. Richa[2] Dr. Rahul Vivek Purohit[3]*

Department of Electronics & Communication Engineering, Galgotias College of Engineering and Technology, Greater Noida, India

[1] panthereagleanmol@gmail.com

[2*] richa.sharma70@gmail.com

[3]puru261280@gmail.com

Abstract

Power quality (PQ) issues, though longstanding, have recently gained significant attention due to increased customer awareness. Among these issues, voltage sags stand out as one of the most critical challenges for electric utilities. This paper examines the key characteristics of voltage sags experienced in power distribution systems. Voltage sag characterization (VSC) is essential for diagnosing, troubleshooting, and implementing effective mitigation strategies. The study emphasizes the significance of two primary sag attributes—magnitude and duration—and also discusses phase-angle jumps caused by faults. MATLAB/SIMULINK is utilized to model and simulate voltage sag scenarios for linear and non-linear loads under symmetrical and asymmetrical fault conditions. The results validate the methodology and offer insights into voltage sag behavior, highlighting its applicability to practical PQ challenges. This detailed analysis provides a foundation for developing robust solutions to mitigate the adverse effects of voltage sags.

Keywords: Power quality (PQ); Voltage sag (VS); Voltage sag characterization (VSC); Simulation

I. INTRODUCTION

The concept of power quality (PQ) has garnered increasing attention from utilities, commercial establishments, and end-users [1,7]. Voltage quality (VQ) plays a pivotal role in PQ, especially for sensitive electrical loads [8] such as medical equipment, industrial machinery, and communication systems. Historically, PQ concerns were centered on reliability, focusing primarily on outages and interruptions. However, modern challenges include more nuanced issues such as voltage sags [2-6], fluctuations, interruptions, and harmonics, which significantly impact operational continuity and equipment longevity [9]. Disaggregation of all parts of the power system is not required in the analysis of voltage sag distribution. These sags are primarily brought about by faults in the system that propagate within the network and may induce hard disturbances. Although voltage sags are less severe than total outages of the power, repetition of voltage sags may initiate serious financial as well as operating consequences. Highly sensitive loads such as adjustable speed drives, programmable logic controllers (PLCs), and data centers are highly susceptible to the disturbances. As a consequence of this, they may incur loss of data, damage to the equipment, or process interruption and hence amplify the overall influence of voltage sags on system operation. In the modeling of voltage sag distributions, no explicit modeling of all components of the power system is necessary. These voltage sags mainly originate from system faults that propagate through the network and thus may result in widespread disturbances. Although voltage sags are generally less harmful than total power outages, their high rate of occurrence can have drastic financial and operational effects. Sensitive loads like adjustable speed drives, programmable logic controllers (PLCs), and data centers are particularly vulnerable to these disturbances. They can lose data, incur equipment damage, or experience process disruption, further increasing the impact of voltage sags on the overall performance of the system. Voltage sag impact mitigation is of paramount importance in ensuring the stability and reliability of power systems. Transient voltage sag can heavily impact sensitive machinery and industrial processes. A proper Voltage Sag Characterization (VSC) forms the foundation for an effective planned voltage sag management. This consists of detection and analysis of leading parameters, i.e., amplitude of the sag (degree of voltage drop), duration (length of time voltage is below the nominal levels), and any accompanying phase-angle oscillations. An understanding of these characteristics is critical to establishing the cause of voltage sags and examining their effect on various equipment in the power system. For example, the magnitude and duration of a sag will dictate whether a specific load can ride through the disturbance without suffering failure. Similarly, phase angle swings can affect the operation of equipment like motors and drives, which are susceptible to such swings. Detailed characterization of voltage sags helps power engineers develop effective mitigation schemes, which include the use of voltage regulators, dynamic voltage restorers (DVRs), and uninterruptible power supplies (UPS). Additionally, it provides crucial data for optimizing network planning and fault management strategies. Overall, these efforts are vital to enhancing power quality, reducing downtime, and improving the reliability of power delivery systems. Through a scientific method of tackling voltage sag problems by characterization and mitigation, industries and utilities are able to save economic losses and ensure uninterrupted process operation [11]. The current document is divided into five separate sections. The first section presents the subject and highlights the challenges of voltage sags. The second section offers an in-depth explanation and description of voltage sags. The third section outlines the theoretical calculations with respect to Voltage Sag Magnitude and Phase-Angle Jump. The fourth section presents the outcomes of the simulation results. Finally, the fifth section integrates the results and presents the conclusions based on the study.

II. VOLTAGE SAG DEFINITION AND CHARACTERIZATION

2.1. Voltage sags (dips)

Voltage sags are a decrease in RMS voltage between 10% and 90% of the normal operating voltage, for a duration between half a cycle and one minute [10]. They are typically due to power system faults, e.g., short circuits, but also from operating high inrush motor loads. For instance, heavy industry processes are expected to produce high inrush currents, which lead to transient voltage drops[1].

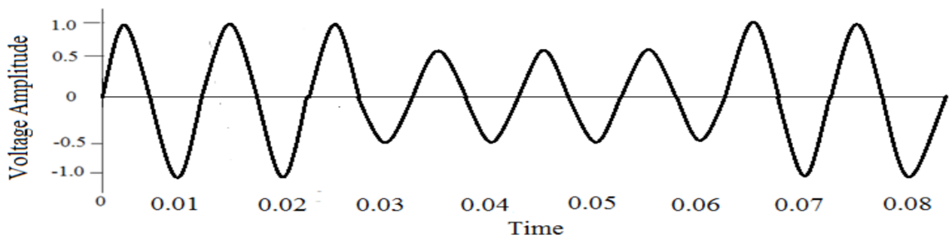


Figure.1 Typical waveform during voltage sag

2.2. Voltage sag characterization (VSC)

Two main features characterize voltage sags:

- **Magnitude**: The magnitude of a voltage sag is the nominal reference voltage minus the voltage at the instant when the sag takes place..This is a parameter of utmost significance in measuring the severity of the disturbance on electrical equipment.
- **Time:** The period of a voltage sag is the time during which the voltage remains below a particular value. It is an important measurement for evaluating possible effects on sensitive loads as well as on system stability overall.

Additionally, voltage sags often cause phase-angle jumps, which represent abrupt changes in the voltage waveform's phase angle[12-16]. These jumps are significant as they affect synchronization in power systems, particularly in systems reliant on precise timing and phase alignment.

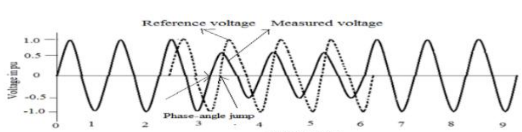


Figure 2 Phase-angle jump + 45°

III. THEORETICAL CALCULATION

To analyze voltage sag phenomena, the voltage divider model is commonly used. This model simplifies the power system by representing it with source and fault impedances, enabling the calculation of sag magnitude and phase-angle shifts. The voltage at the point of common coupling (PCC) can be expressed as:

Where: Z_s is the source impedance, Z_f is the fault impedance and V_s is the source voltage.

The voltage at the point of common coupling (PCC) is calculated using the voltage divider rule, which considers the relationship between the source impedance (Z_s) and fault impedance (Z_f). It is given by:

$$V_{pcc} = V_s \times \frac{Z_f}{Z_s + Z_f} \qquad (1)$$

This equation helps determine the sag magnitude and phase-angle variations based on system impedance conditions. The PAJ is a crucial parameter that provides insights into the dynamic behavior of the power system during faults, affecting equipment synchronization and stability.

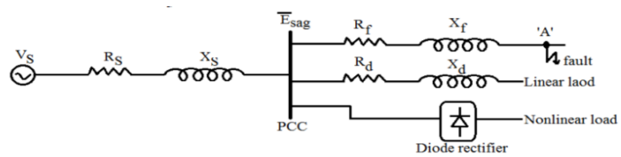


Figure. 3 Voltage divider model for voltage sag magnitude [1]

In order to examine the origin of voltage-sag phase-angle jump (PAJ), the single-phase voltage divider model of Figure 2 can be utilized. This model considers the source impedance (Z_s) and fault impedance (Z_f) as complex values [17]. Using per-unit calculation, by neglecting all load currents. Voltage at the point of common coupling (PCC).

$$\bar{E}_{sag} = \left[\left(R_f + jX_f\right) \Big/ \left(R_s + jX_s\right) + \left(R_f + jX_f\right) \right] \bar{V} \qquad (2)$$

If V (avg. voltage) =1, then equation (1) is rewritten as

$$\bar{E}_{sag} = \left[\left(R_f + jX_f\right) \Big/ \left(R_s + jX_s\right) + \left(R_f + jX_f\right) \right] \qquad (3)$$

The argument of $\bar{E}_{sag}$, equivalent to the phase-angle jump (PAJ) is given by the following expression

$$\Delta\emptyset = \arg\left(\bar{E}_{sag}\right) = \arctan\left(\frac{X_f}{R_f}\right) - \arctan\left(\frac{X_f + X_s}{R_f + R_s}\right) \qquad (4)$$

where R_s, X_s, R_f, X_f and $\bar{E}_{sag}$ is the source resistance, source reactance, feeder resistance, feeder reactance and voltage sag at the point of common coupling (PCC)[18-19].

IV SIMULATION RESULTS

The proposed system shown in Fig 4 is a Simulink-based simulation environment designed to evaluate the characteristics of voltage sag in a power distribution network under different fault conditions. This system gives a comprehensive modeling, simulation, and evaluation of the effects of symmetrical and asymmetrical faults on voltage magnitude, phase-angle, and duration. The setup is an 11 kV, 30 MVA, 50 Hz three-phase voltage source for a high-voltage distribution feeder. A delta/wye transformer (11 kV/0.4 kV, 1 MVA) is used to step down the voltage, thus making it a practical setup for studying sag propagation from high voltage to low voltage levels. To simulate real load conditions and their response to faults, a 10 kW resistive load and a 100 VAR inductive load are used. Two fault blocks are included to permit a range of faults, including:

- Single-line-to-ground faults
- Double-line-to-ground faults
- Line-to-line faults
- Three-phase faults
- Multistage faults

Fault events are simulated in the model to include dynamic responses. The system describes voltage sags in terms of magnitude, duration, and phase-angle jump and accurately records the following fault responses:

- Propagation of voltage sags across transformer windings
- Asymmetry in sag due to transformer and load settings

The designed system acts as a key facility for power quality analysis that supports thorough researches of voltage disturbances to boost power system reliability and efficiency.

4.1 Line-to-Line Faults

The A-B line-to-line fault between phases simulation indicated that the amplitude of the sag is dependent on fault resistance and transformer configuration. Phases A and B experienced large voltage sags at the 11 kV bus, while phase C experienced small swells. The results indicate the impact of transformer configuration and fault impedance on sag propagation. This line fault model is capable of simulating multiple fault types, including:

- Single-line-to-ground faults
- Double-line-to-ground faults
- Line-to-line faults
- Three-phase faults
- Multistage faults

Voltage sag waveforms initiated by a phase-to-phase fault between phases A and B on an 11 kV feeder line between 0.1 and 0.2 seconds are shown in Figures 5 and 6. As can be observed, the 11 kV bus experiences two totally different voltage sags at phases A and B with different sag magnitudes. The cause of the difference is the high fault resistance of 8 Ω between the two faulted phases. As the voltage sag propagates downstream through the 11 kV/0.4 kV, 1 MVA transformer to the load, the fault type [15] is altered due to the delta/wye transformer configuration. Additionally, the unfaulted phase C at the 0.4 kV bus experiences a slight voltage swell, which occurs due to the absence of a ground point in the line-to-line fault and the presence of high fault resistance

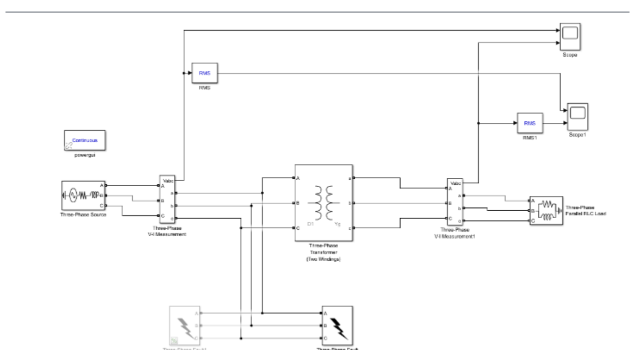


Figure. 4 Line fault Simulink model

4.2 Multistage Faults

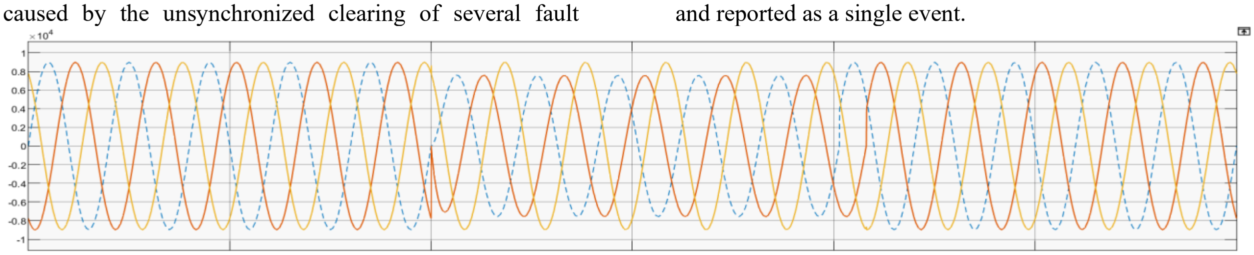
Multistage voltage sags were also investigated, where the actions of asynchronous relays or fault impedances caused a series of multistage sags before the system returned to nominal voltage. Multistage sags are important for distributed generation systems or systems with sophisticated protection systems.In voltage sag waveform amplitude, it is usually indicated in RMS terms and normalized to make it clearly visible.

4.3 RMS Analysis

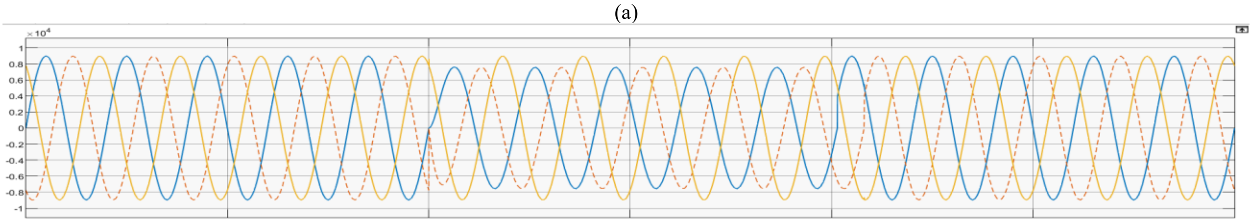
Instantaneous and RMS waveforms for every fault case were developed. RMS analysis normalized voltage sag waveforms to provide a more realistic indication of sag magnitudes and durations. Pre-sag and post-sag time interval oscillations were induced by transient phase shifts during the fault occurrences.

Figures 7 and 8 illustrate the RMS plot of sag waveforms of the line-to-line fault voltage in Figures 5 and 6.

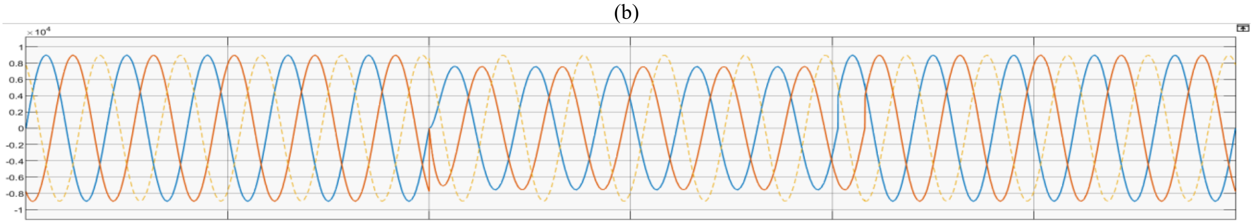
The amplitudes of the sag for all of the phases are evidently illustrated. The slight oscillation seen before and after the sag and swell is a result of the phase shift during the fault. The line fault model can simulate faults that occur in several stages. Voltage sags occurring in several stages are normally caused by the unsynchronized clearing of several fault protection relay mechanisms, which lead to variations in the power system's impedance and network configuration. This results in a sequence of voltage sags before the nominal level is restored [20], or due to fault or ground impedance variations at the occurrence of a fault. It is at times referred to as a sequence of faults in close chronological proximity and reported as a single event.



(a)

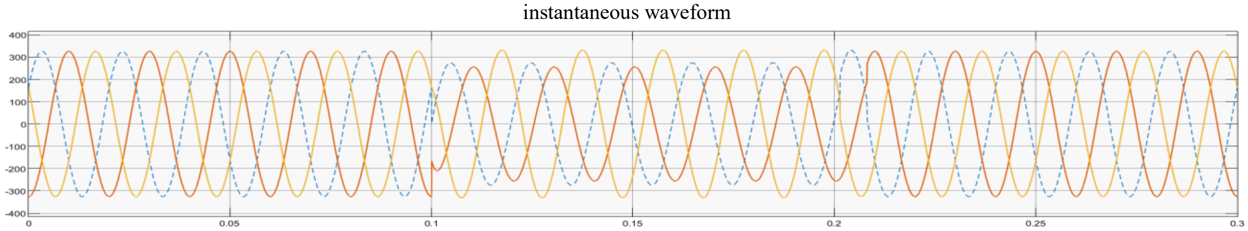


(b)

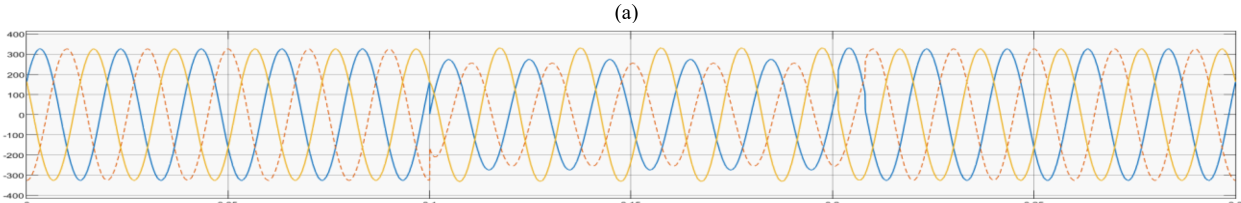


(c)

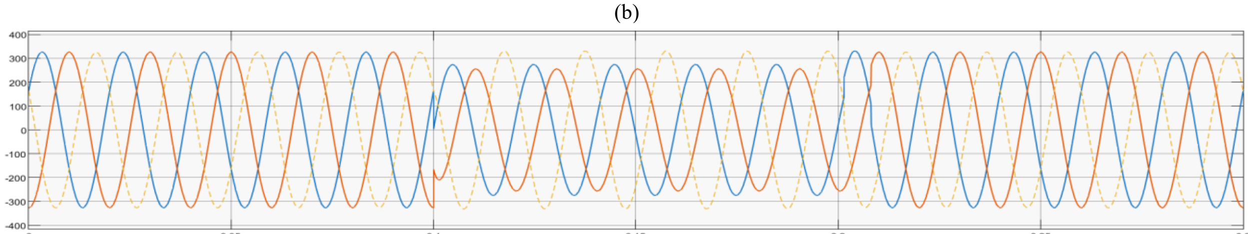
Figure. 5 (a) by phase a (b) by phase b (c) by phase c illustrates the voltage sag caused by a line-to-line fault at the 11 kV bus, represented as an instantaneous waveform



(a)

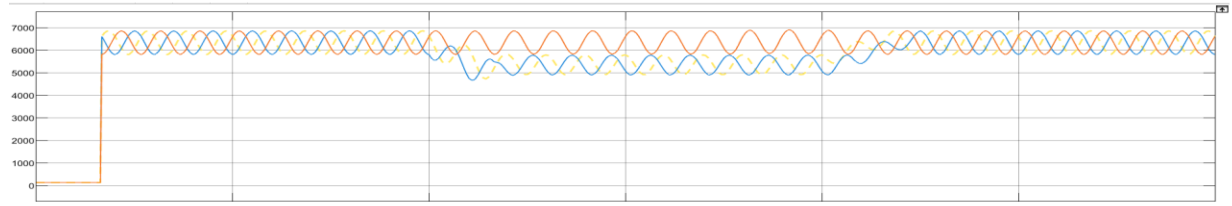


(b)

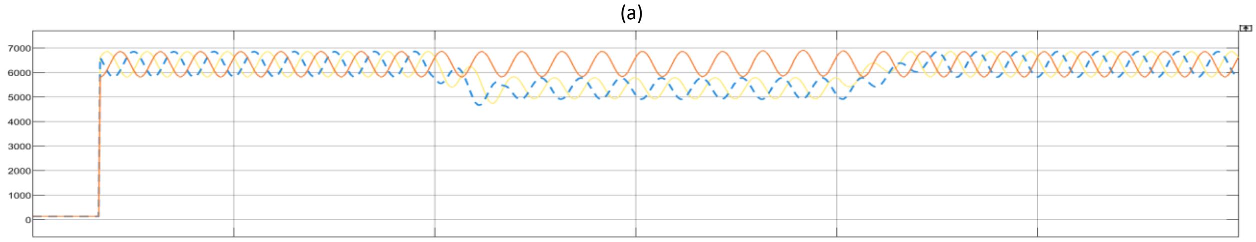


(c)

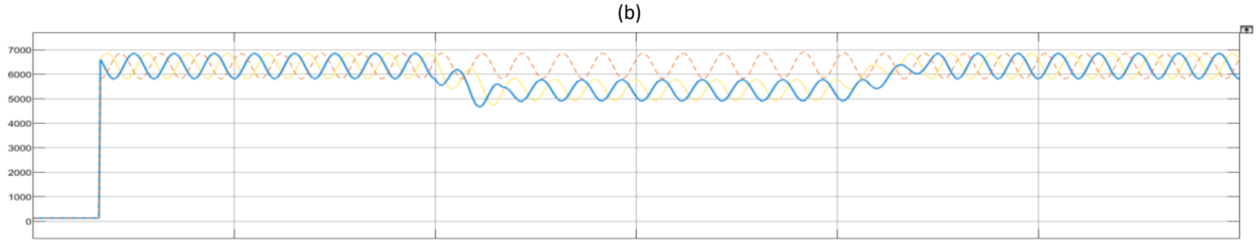
Figure 6 (a) by phase a (b) by phase b ci) by phase c illustrates the voltage sag caused by a line-to-line fault at the 0.4 kV bus, represented as an instantaneous waveform.



(a)

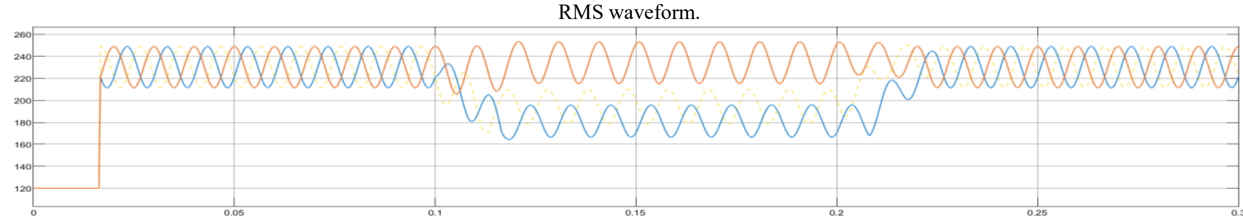


(b)

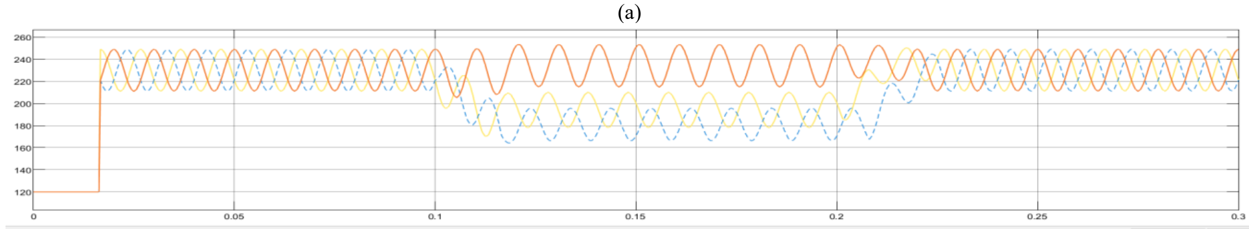


(c)

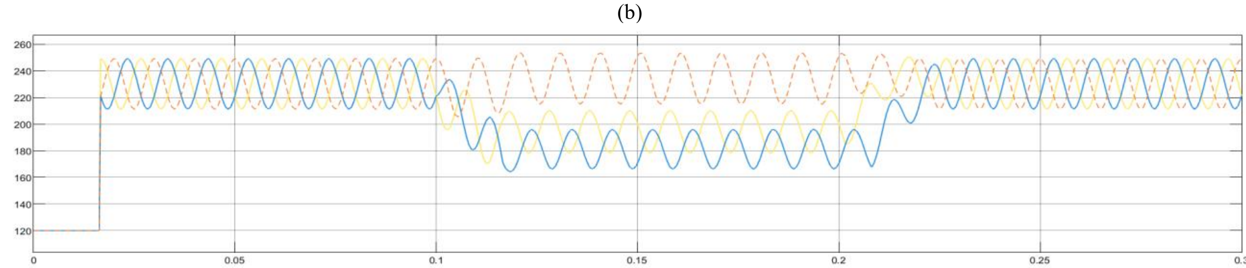
***Figure7(*a)** by phase a (b) by phase b (c) by phase c illustrates the voltage sag caused by a line-to-line fault at the 11 kV bus, represented as an RMS waveform.



(a)



(b)



(c)

Figure 8(a) by phase a (b) by phase b (c) by phase c illustrates the voltage sag caused by a line-to-line fault at the 0.4 kV bus, represented as an instantaneous waveform.

V Conclusion

This study provides an in-depth analysis of voltage sag characteristics, a significant PQ issue affecting modern power systems. The research emphasizes the importance of detailed voltage sag characterization, including magnitude, duration, and phase-angle jumps, for developing effective mitigation strategies. Using MATLAB/SIMULINK, the study successfully simulated various fault scenarios, demonstrating the robustness of the proposed methodology. The findings underscore the need for advanced diagnostic tools and strategies to address voltage sag challenges in both three-phase and single-phase systems.

Future work could focus on integrating renewable energy sources and distributed generation into voltage sag studies to evaluate their impact on sag characteristics and mitigation techniques.

REFERENCES

1.M. Prasad, A. K. Akella "Voltage sag characteristics in power distribution system under fault conditions" Energy Education Science and Technology Part A: Energy Science and Research 2015 Volume (issue) 33(6): 3177-3192

2.Rodney H.G. Tan and Vigna K. Ramachandaramurthy "A Comprehensive Modeling and Simulation of Power Quality Disturbances Using MATLAB/SIMULINK"

3.Gulali Yalcinkaya, Member, IEEE, Math H. J. Bollen, Senior Member, IEEE, and Peter A. Crossley, Member, IEEE "Characterization of Voltage Sags in Industrial Distribution Systems" (JULY/AUGUST 1998)

4.Pirjo Heine, Member, IEEE, and Matti Lehtonen "Voltage Sag Distributions Caused by Power System Faults" IEEE TRANSACTIONS ON POWER SYSTEMS, VOL. 18, NO. 4, NOVEMBER 2003

5.Saeed Lotfifard, Student Member, IEEE, Mladen Kezunovic, Fellow, IEEE and Mirrasoul J. Mousavi, Member, IEEE "Voltage Sag Data Utilization for Distribution Fault Location" IEEE TRANSACTIONS ON POWER DELIVERY, VOL. 26, NO. 2, APRIL 2011

6.Miska Prasad, and A.K Akella "Comparison of Voltage Swell Characteristics in Power Distribution System" International Journal of Electrical & Electronics Research (IJEER) Volume 4, Issue 3, Pages 67-73, September 2016

7.Babaei E, Kangarlu MF. Sensitive load voltage compensation agaist voltage sags/swells and harmonics in the grid voltage and limit downstream fault currents using DVR. Electr Power Syst Res 2012;83:80-90

8. Babaei E, Kangarlu MF. Cross-phase voltage compensator for three phase distribution systems, Electr Power Energ Syst 2013;53:119-126

9. Alam MR, Muttaqi KM, Bouzerdoum A. Characterizing voltage sags and swells using three-phase voltage Ellipse parameters. IEEE Trans Indust Appl 2015;51:2780-2790

10. IEEE Standards Coordinating Committee 22 on Power Quality. IEEE Std. 1159. IEEE Recommended Practice for Monitoring Electric Power Quality, 1995

11. Piumetto M, Gomez Targarona JC. Characterization of voltage sags and its impact on sensitive loads in a MV system with distributed generation for single-phase fault. IEEE Latin America Trans FEB 2013;11:439-446.

12. G. Yal¸cinkaya and M. H. J. Bollen, "Stochastic assessment of voltage sags for systems with large induction motor loads," presented at the Universities Power Engineering Conf., Galway, Ireland, Sept. 1994

13. C. Guo, X. Zhang and L. Wang, "Economic Evaluation of Voltage Sag under Uncertain Conditions Based on Beta Distribution Parameter Correction," *2023 IEEE International Conference on Power Science and Technology (ICPST)*, Kunming, China, 2023, pp. 917-922, doi: 10.1109/ICPST56889.2023.10165173.

14. K. Ghambirlou, A. A. Abosnina and G. Moschopoulos, "Voltage Sag Investigation for Induction Motor Startup in Distribution Systems using Solid State Transformer," *2025 15th International Renewable Energy Congress (IREC)*, Hammamet, Tunisia, 2025, pp. 1-6, doi: 10.1109/IREC64614.2025.10926753.

15. X. Wu, H. Zhong, Y. Chen and Y. Hu, "A Novel Voltage Sag Detection Method for Series Filter Originate from Improved Genetic Algorithm," *2024 International Conference on Intelligent Communication, Sensing and Electromagnetics (ICSE)*, Guangzhou, China, 2024, pp. 269-274, doi: 10.1109/ICSE63502.2024.10899052.

16. Hadi S (2004) Power System Analysis, 2nd edn, McGraw Hill.

17. Bollen MH. Understanding power quality problems: Voltage sags and interruptions. Wiley-IEEE Press, p. 672, September 1999.

18. Ms.Minakshi V.Bhakkad and Prof.B.T.Deshmukh, "Generation of Voltage Sag for different loads and conditions using MATLAB SIMULINK" 2019 International Conference on Innovative Trends and Advances in Engineering and Technology (ICITAET)

19. Juan Ram´ on Camarillo-Pe˜ naranda and Gustavo Ramos, "Effect of phase-angle jump in Impedance-based Fault Location Methods for Transmission and Distribution Systems" 2018 IEEE IAS annual meeting.

20. G. Yal¸cinkaya and M. H. J. Bollen, "Stochastic assessment of frequency, magnitude and duration of nonrectangular voltage sags in a large industrial distribution system," in Proc. 12th Power System Computation Conf., Dresden, Germany, Aug. 19–23, 1996, vol. II, pp. 1018–102